

Indirect Estimation of Passive-Balancing Current from Two Repeated Charge Cycles

Differential matched-state method for the Dyness Powerbrick Pro 14.3 kWh BMS

Battery software version 2.50-71.10.11

Technical calculation note - 9 August 2026

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Scope. This note documents an indirect engineering calculation from recorded telemetry. It is not an official Dyness specification, does not replace manufacturer limits, and does not establish permission to charge against a BMS command. The 3.6 V cell level is treated only as the observed protection boundary, not as a normal charge target.

Abstract

Two repeated low-current charge cycles were used to estimate the current of the internal passive-balancing channels without opening the battery or measuring a resistor directly. Cell 7 was the highest cell in both cycles and was used as a state marker. At the same charging current of 1.5 A, the two cycles were matched at cell-7 voltages of 3.500 V and 3.599 V. The second cycle showed pack voltages higher by 30 mV and 60 mV, respectively, while cell 7 itself was at the same voltage. Under the stated assumption that the same cell at the same current, temperature region and voltage is at the same state of charge, the other cells were therefore relatively more charged after one balancing loop.

The relative charge displacement of every directly transmitted cell was obtained from its local differential capacity. It was then equated to the difference of the integrated bleed-current exposure between that cell and cell 7. A pooled least-squares fit gives an effective bleed resistance of 58.7–60.7 Ω , depending on treatment of reconstructed cell 16 in the minimum-cell threshold. The midpoint is 59.7 Ω , corresponding to approximately 58.6–60.3 mA between 3.5 V and 3.6 V. The rounded engineering estimate is therefore **about 60 mA per selected cell**, with a conservative interpretation range of roughly 50–70 mA.

Result at a glance

Effective bleed resistance	approximately 60 Ω
Current at 3.5 V	approximately 59 mA
Current at 3.6 V	approximately 60 mA
Power per active channel near the top	approximately 0.20–0.22 W
Bleed charge removed from cell 7 per matched loop	approximately 0.068 Ah
Conservative engineering range	50–70 mA

1. Measurement basis and matched reference states

1.1 Source data and non-duplication

The calculation uses the telemetry files named in Appendix B. The second CSV contains 3,059 samples and includes the entire 1,448-row content of the first CSV as an exact parsed prefix. Consequently, the second file alone was used for the two-cycle calculation; the first file was used only as a consistency check and was not added a second time. The median sample interval is 6 s, with observed intervals of 5–7 s.

The logger records all 16 physical cells. Cells 1–15 are transmitted directly. Cell 16 is reconstructed as

$$V_{16} = V_{\text{pack}} - \sum_{i=1}^{15} V_i. \quad (1)$$

The reconstruction is physically useful but inherits pack-voltage quantization and the accumulated quantization of cells 1–15. For millivolt-scale threshold timing, the direct cells therefore provide the higher-resolution treatment.

The balancing rule used in the calculation is the experimentally inferred rule from the associated report:

$$\text{resistor } i \text{ selected when } V_i - V_{\min} \geq 30 \text{ mV}, \quad (2)$$

provided that the global balancing gate is enabled by charging current. The selection threshold is a cell-voltage difference of 30 mV.

1.2 Reference-cell assumption

Cell 7 is the maximum-voltage cell at both upper reference points. The calculation assumes:

For the same physical cell, at the same charging current and approximately the same temperature, the same terminal voltage represents the same cell state of charge.

Therefore, when cell 7 is observed at the same voltage in both cycles, its charge-state coordinate is taken to be equal in the two records. This converts cell 7 into an internal state marker and permits a differential comparison that does not require the absolute capacity or absolute SOC to be known.

Because the 16 cells are series-connected,

$$V_{\text{pack}} = \sum_{i=1}^{16} V_i, \quad (3)$$

$$\Delta V_{\text{pack}} = \sum_{i=1}^{16} \Delta V_i. \quad (4)$$

At a matched cell-7 state, $\Delta V_7 = 0$. A positive ΔV_{pack} therefore directly means that the sum, and hence the average, of the other 15 cell voltages is higher in the second cycle.

1.3 Observed matched states

Table 1. Matched cell-7 states used in the calculation. Both cycle values are recorded at 1.5 A charging current.

Cell-7 anchor	Cycle 1 sample/time	$V_{\text{pack},1}$ (V)	Cycle 2 sample/time	$V_{\text{pack},2}$ (V)	ΔV_{pack} (mV)	I (A)
3.500 V	965 / 16:05:01	54.51	2329 / 18:21:26	54.54	+30	1.5
3.599 V	1103 / 16:18:49	55.05	2467 / 18:35:15	55.11	+60	1.5

The 3.599 V anchor is the last recorded telemetry level immediately below the nominal 3.600 V charge-MOSFET cutoff. At the 3.500 V reference, the BMS temperature was 39.55 °C in cycle 1 and 40.15 °C in cycle 2; ordinary cell-temperature sensors differed by only about 0.1–0.2 °C. At the 3.599 V reference, the BMS temperatures were 40.65 °C and 40.85 °C. These differences are small relative to the indirect nature of the method.

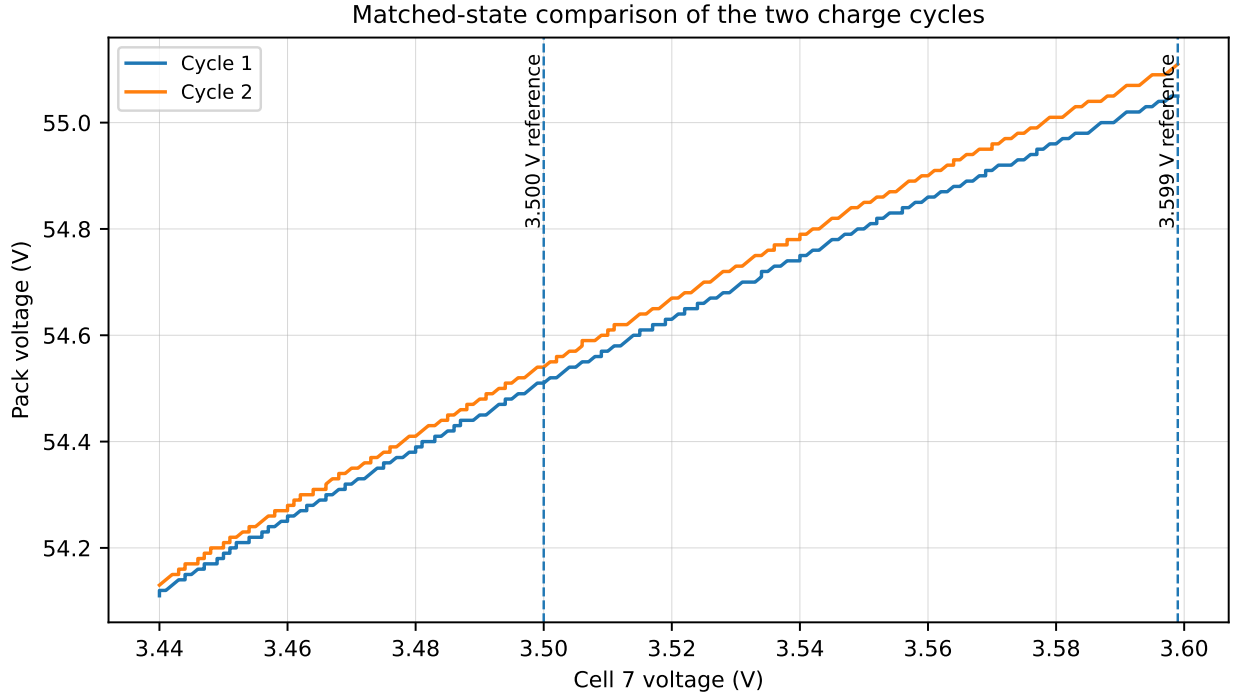


Figure 1. Pack voltage plotted against the reference-cell voltage in the upper charge region. At the same cell-7 voltage, the second cycle lies above the first cycle. The separation is approximately 30 mV at 3.500 V and 60 mV at 3.599 V.

2. Differential charge-balance model

2.1 Balancer-selection function

Let $V_i(t)$ be the terminal voltage of cell i , and let

$$V_{\min}(t) = \min_{1 \leq i \leq 16} V_i(t). \quad (5)$$

Let $G(t) \in \{0, 1\}$ denote the global balancer gate, and let $H(\cdot)$ be the Heaviside step function. The inferred individual selection state is

$$b_i(t) = G(t) H[V_i(t) - V_{\min}(t) - 30 \text{ mV}]. \quad (6)$$

For the recorded loops, $G(t)$ is represented by charging current at or above 1.5 A. Any short qualification delay has negligible effect here because the 30 mV cell spread is reached long after the stable charge current begins.

The balancing channel is represented by a common effective resistance R_{eff} . This parameter includes the physical resistor, switching resistance and any unobserved pulse-width modulation or duty cycle. The effective bleed current is

$$I_{b,i}(t) = b_i(t) \frac{V_i(t)}{R_{\text{eff}}}. \quad (7)$$

2.2 Integrated bleed charge

For an interval from t_a to t_b , define

$$K_i(t_a, t_b) = \int_{t_a}^{t_b} b_i(t) V_i(t) dt, \quad (8)$$

$$B_i(t_a, t_b) = \int_{t_a}^{t_b} I_{b,i}(t) dt = \frac{K_i(t_a, t_b)}{R_{\text{eff}}}. \quad (9)$$

With time expressed in hours, K_i has units of volt-hours and B_i has units of ampere-hours.

Let q_i be a cell charge-state coordinate in ampere-hours. Neglecting cell-specific leakage, the integrated charge balance of a series-connected cell is

$$q_i(t_b) - q_i(t_a) = \int_{t_a}^{t_b} I_{\text{bat}}(t) dt - \frac{K_i(t_a, t_b)}{R_{\text{eff}}}. \quad (10)$$

The series-current integral is common to all cells. Subtracting the equation for reference cell $r = 7$ from the equation for cell i gives

$$\Delta q_i - \Delta q_7 = \frac{K_7 - K_i}{R_{\text{eff}}}. \quad (11)$$

At matched cell-7 states, the central assumption is $\Delta q_7 = 0$. Hence

$$\boxed{\Delta q_i = \frac{K_7 - K_i}{R_{\text{eff}}}.} \quad (12)$$

This is the key cancellation. Discharge, recharge, a common BMS load and a constant pack-current offset affect the common series-current term, but do not appear in Equation (12). Only cell-specific effects remain.

3. Conversion of voltage displacement into charge displacement

3.1 Local differential capacity

At fixed charging current and temperature region, the terminal voltage of a cell is treated locally as a single-valued function of its charge state,

$$V_i = F_i(q_i, I, T). \quad (13)$$

The observed voltage displacement between the two matched cycles is

$$\Delta V_{i,k} = V_{i,k}^{(2)} - V_{i,k}^{(1)}, \quad (14)$$

where k denotes either the 3.500 V or 3.599 V reference state of cell 7. The corresponding charge displacement is, in general,

$$\Delta q_{i,k}^{\text{obs}} = \int_{V_{i,k}^{(1)}}^{V_{i,k}^{(2)}} C_{d,i}(V) dV, \quad C_{d,i}(V) = \frac{dq_i}{dV_i}. \quad (15)$$

During a local constant-current charge segment,

$$C_{d,i} \approx \frac{I_{\text{ref}}}{\dot{V}_i}, \quad \dot{V}_i = \frac{dV_i}{dt}. \quad (16)$$

The slope $\dot{V}_i$ is cell-specific, so different cell capacities and different positions on the LiFePO_4 charge curve are handled locally rather than by assuming a generic voltage-SOC curve.

3.2 Implemented local approximation

For each cell, each cycle and each reference point, $\dot{V}_i$ is estimated by ordinary least squares over the eight minutes immediately preceding the reference sample, restricted to 1.4–1.6 A. If $c \in \{1, 2\}$ denotes the cycle,

$$\hat{V}_{i,k}^{(c)} = \frac{\sum_j (t_j - \bar{t})(V_{i,j} - \bar{V}_i)}{\sum_j (t_j - \bar{t})^2}. \quad (17)$$

The differential capacities from the two cycles are averaged:

$$\bar{C}_{d,i,k} = \frac{I_{\text{ref}}}{2} \left(\frac{1}{\hat{V}_{i,k}^{(1)}} + \frac{1}{\hat{V}_{i,k}^{(2)}} \right). \quad (18)$$

For the small observed voltage displacement, Equation (15) is then approximated by

$$y_{i,k} \equiv \Delta q_{i,k}^{\text{obs}} \approx \bar{C}_{d,i,k} \Delta V_{i,k}. \quad (19)$$

An iterative correction replacing I_{ref} with $I_{\text{ref}} - b_i V_i / R_{\text{eff}}$ changes the fitted resistance by only about one to two percent, smaller than the slope-window and quantization sensitivity. The simpler form above is therefore retained as the primary transparent calculation.

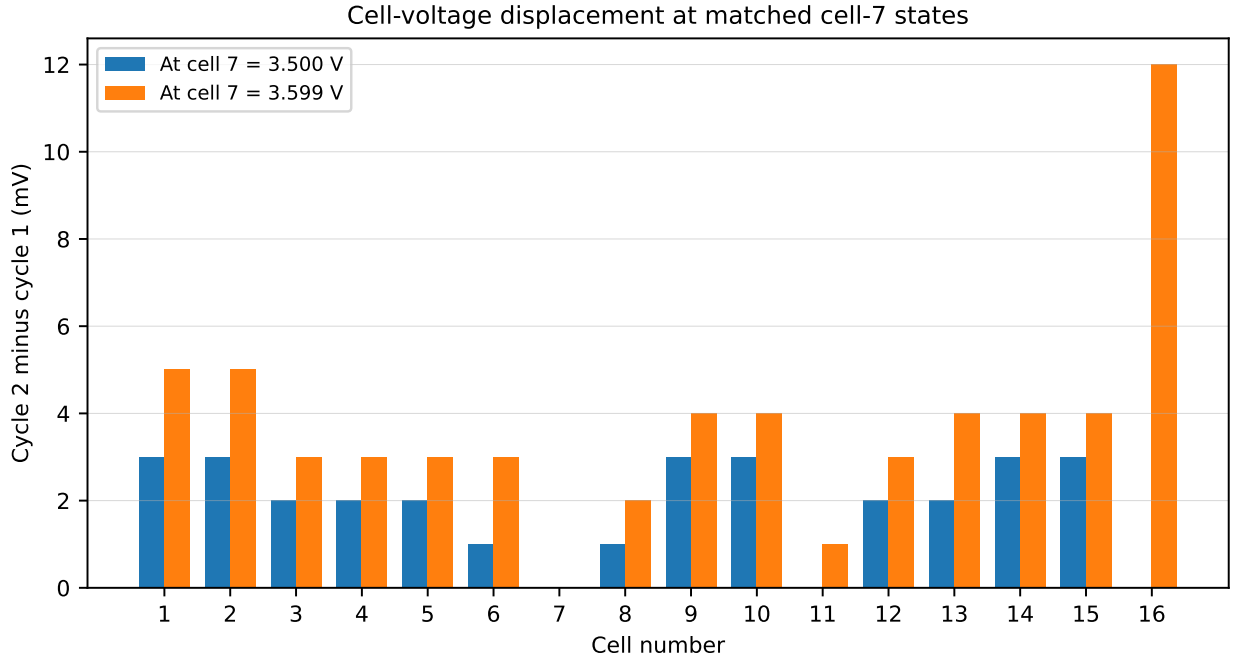


Figure 2. Cell-voltage displacement between cycle 2 and cycle 1 at the two matched cell-7 states. Cell 7 is fixed by construction. Cell 16 is reconstructed and shows the largest high-anchor quantization-sensitive displacement; it is not used as a regression observation.

4. Discrete integration and resistance identification

4.1 Trapezoidal evaluation of the bleed integrals

For telemetry samples $j = m, \dots, n$ with time in hours, Equation (8) is evaluated using the trapezoidal rule:

$$K_i \approx \sum_{j=m}^{n-1} \frac{b_{i,j} V_{i,j} + b_{i,j+1} V_{i,j+1}}{2} (t_{j+1} - t_j). \quad (20)$$

The inferred resistor-on time is evaluated analogously:

$$T_i \approx \sum_{j=m}^{n-1} \frac{b_{i,j} + b_{i,j+1}}{2} (t_{j+1} - t_j). \quad (21)$$

Two threshold treatments are retained:

- **Direct-channel treatment:** $V_{\min}$ is calculated from cells 1–15 to avoid reconstructed-cell quantization in millivolt-scale switching times.
- **All-cell treatment:** $V_{\min}$ is calculated from all 16 physical-cell representations, matching the intended physical algorithm but allowing reconstructed cell 16 to influence threshold timing.

In both cases, only directly transmitted cells 1–15 are used as fit observations. Cell 7 is the reference and is excluded. A zero-millivolt displacement that is unresolved at the telemetry resolution is also excluded from the fit.

4.2 Least-squares estimator

For each usable cell i and reference point k , define

$$x_{i,k} = K_{7,k} - K_{i,k}. \quad (22)$$

Equations (12) and (19) give the linear model

$$y_{i,k} = \frac{x_{i,k}}{R_{\text{eff}}} + \varepsilon_{i,k}. \quad (23)$$

Writing $a = 1/R_{\text{eff}}$ and forcing the physically required zero intercept, the unweighted least-squares estimate is

$$\hat{a} = \frac{\sum_{i,k} x_{i,k} y_{i,k}}{\sum_{i,k} x_{i,k}^2}, \quad (24)$$

$$\widehat{R_{\text{eff}}} = \frac{\sum_{i,k} x_{i,k}^2}{\sum_{i,k} x_{i,k} y_{i,k}}. \quad (25)$$

The two reference levels are pooled in one fit. Separate fits are also reported as a consistency check.

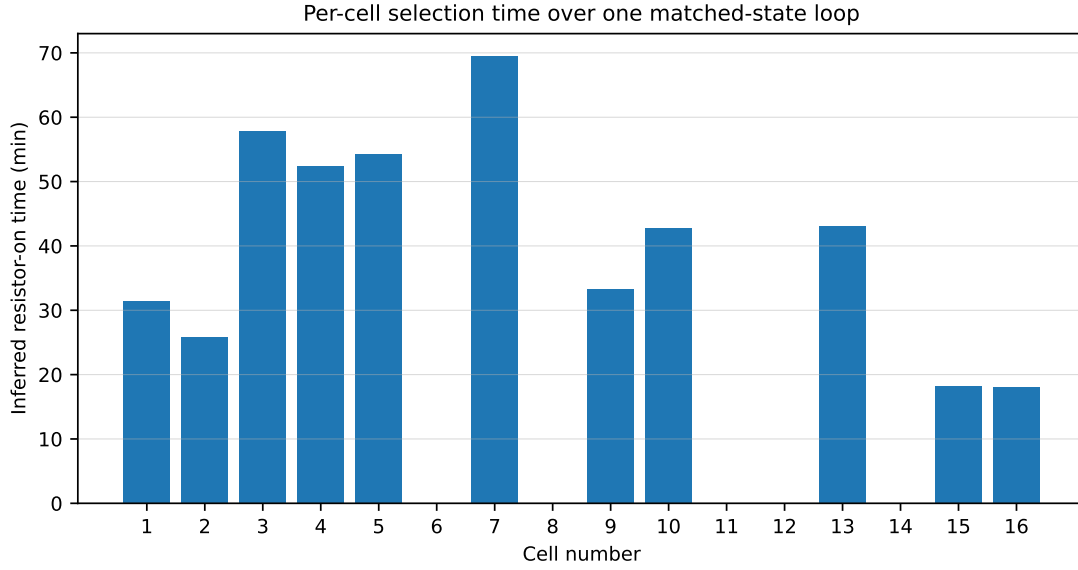


Figure 3. Inferred per-cell resistor-on time for one matched-state loop using the direct-channel minimum. Cell 7 is selected for approximately 69.5 min; several low cells are never selected.

5. Numerical results

5.1 Integrated exposure of the reference cell

The two matched-state intervals produce almost identical exposure integrals, as expected for two descriptions of one complete loop:

Table 2. Reference-cell balancing exposure and net pack charge over the matched loops.

Treatment	Cell-7 on-time (min)	K_7 (V h)	Fitted R_{eff} (Ω)	Cell-7 bleed charge (mAh)
Direct cells 1–15 define V_{min}	69.5	3.989	58.7	67.9
All 16 cells define V_{min}	71.4	4.095	60.7	67.5

The independently obtained reference-cell bleed charges, 67.9 mAh and 67.5 mAh, differ by less than one percent. This agreement is stronger than the difference in inferred switching time because the fitted resistance changes consistently with the threshold treatment.

The measured net pack-current integrals are 0.175 Ah for the 3.500 V loop and 0.176 Ah for the 3.599 V loop. These values are deliberately not used as the balancing charge because they include common-mode contributions that the differential method cancels.

5.2 Fitted effective resistance and current

Table 3. Least-squares resistance estimates from the two threshold treatments. Current values use $I = V/R_{\text{eff}}$.

Threshold treatment	Pooled R_{eff} (Ω)	R_{eff} at 3.500 V (Ω)	R_{eff} at 3.599 V (Ω)	I_b at 3.5 V (mA)	I_b at 3.6 V (mA)
Direct cells 1–15 define $V_{\min}$	58.7	60.2	57.5	59.6	61.3
All 16 cells define $V_{\min}$	60.7	62.2	59.4	57.7	59.3

The midpoint of the two pooled treatments is $59.7\ \Omega$. At this value,

$$I_b(3.5\ \text{V}) = \frac{3.5\ \text{V}}{59.7\ \Omega} = 58.6\ \text{mA}, \quad (26)$$

$$I_b(3.6\ \text{V}) = \frac{3.6\ \text{V}}{59.7\ \Omega} = 60.3\ \text{mA}, \quad (27)$$

$$P_b(3.5\ \text{V}) = \frac{V^2}{R_{\text{eff}}} = 0.205\ \text{W}, \quad (28)$$

$$P_b(3.6\ \text{V}) = \frac{V^2}{R_{\text{eff}}} = 0.217\ \text{W}. \quad (29)$$

The rounded engineering representation is therefore a $60\ \Omega$ effective channel drawing about 60 mA near the upper balancing region.

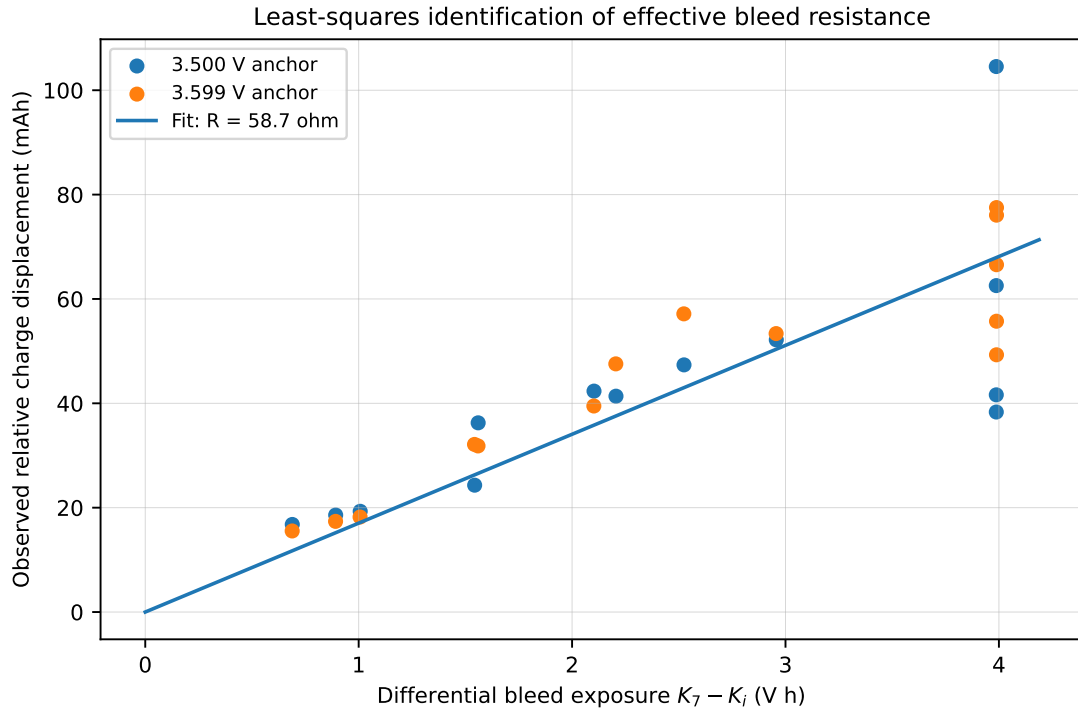


Figure 4. Pooled least-squares fit for the primary high-resolution threshold treatment. The through-origin coefficient of determination is $R_0^2 = 0.934$ and the residual root-mean-square value is 12.5 mAh. The scatter is expected from 1 mV cell-voltage quantization and local charge-curve curvature.

6. Sensitivity and uncertainty

6.1 Slope-window sensitivity

The local differential capacity is the largest model-dependent element because the observed voltage changes are only a few millivolts. Repeating the complete fit with local regression windows from 5 to 12 minutes gives resistance estimates from $55.7\,\Omega$ to $63.2\,\Omega$. An eight-minute window is used as a compromise: it contains roughly 80 samples, suppresses 1 mV quantization noise, and remains local relative to the steep upper-charge curve.

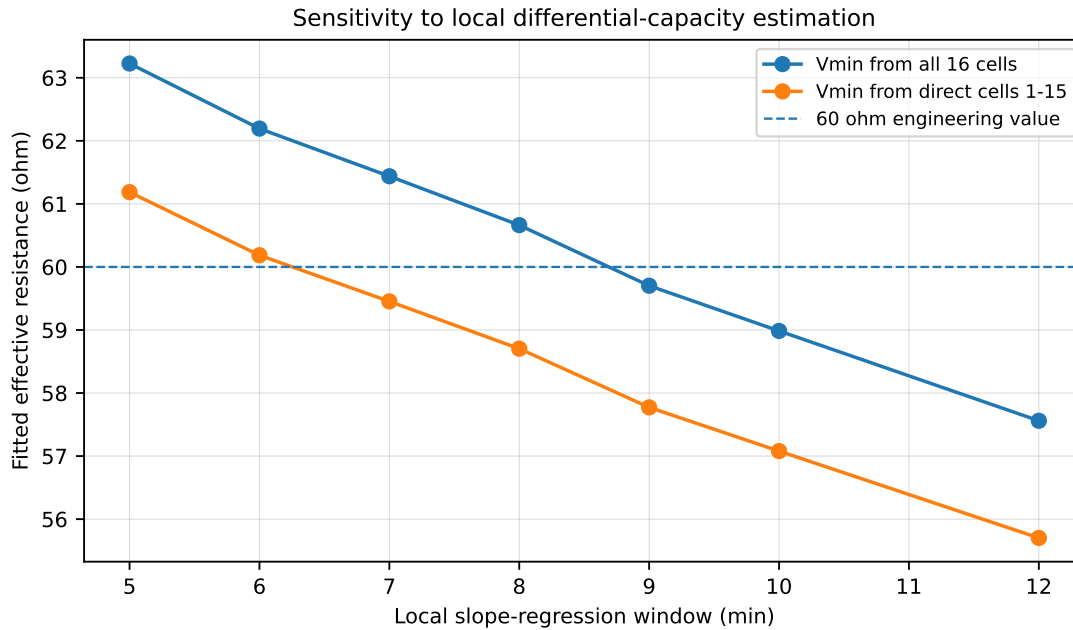


Figure 5. Sensitivity of the pooled resistance estimate to the local slope-regression window. The rounded $60\,\Omega$ value lies near the center of the plausible treatments.

A leave-one-cell-out check gives $56.6\text{--}64.7\,\Omega$. These ranges are not formal statistical confidence intervals because observations from the same physical cells and the two anchors are correlated. They are engineering sensitivity checks against modeling choices and influential cells.

6.2 Why direct pack-current integration is not the estimator

A tempting calculation is to divide the net pack-current integral, approximately $0.175\,\text{Ah}$, by the roughly $1.17\,\text{h}$ cell-7 selection time. That produces an apparent current around $0.15\,\text{A}$, much higher than the differential estimate. The pack integral, however, contains all common terms: BMS electronics consumption, current-sensor offset, charge/discharge inefficiency and any background load. The CSV cannot separate those terms from balancing current.

Equation (11) removes common series-current contributions by subtracting one cell from another. This is why the relative-cell method is preferred for estimating the resistor current.

6.3 Interpretation of the reported current

The reported value is an *effective average current while a cell is selected*. If the physical circuit uses pulse-width modulation, intermittent scanning or a resistor plus semiconductor drops, the fitted R_{eff} represents their combined average effect. It should not automatically be interpreted as the DC resistance

of a single discrete resistor measured with an ohmmeter.

7. Limitations

The calculation depends on the following assumptions and limitations:

- Equal cell voltage at equal charging current and close temperature is treated as equal cell SOC. Voltage hysteresis or relaxation differences can violate this approximation.
- The balancer-selection state is inferred from the 30 mV rule; no explicit per-cell balancing flag is available in the decoded telemetry.
- Cell voltages are quantized to approximately 1 mV; most matched-state displacements are only 1–5 mV.
- Local differential capacity is estimated from a finite regression window. The resulting window sensitivity is explicitly included in the uncertainty assessment.
- Cell 16 is reconstructed from pack voltage and cells 1–15. It remains part of the physical pack model but is excluded as a fit observation.
- Cell-specific leakage currents or electronics powered asymmetrically from cell taps would not cancel perfectly and are not separately measured.
- The result characterizes one tested Dyness Powerbrick Pro battery and software version 2.50-71.10.11. Other revisions may differ.
- No direct resistor-current probe or calorimetric calibration was performed. The estimate is therefore indirect and should be reported as an engineering inference.

8. Conclusion

Under the stated matched-state assumption, the two repeated charge cycles support an effective passive-balancing resistance of approximately $60\ \Omega$. This corresponds to about 58–61 mA per selected cell between 3.5 V and 3.6 V, with approximately 0.20–0.22 W dissipated per active channel. Cell 7 lost about 0.068 Ah through its balancing channel over one complete matched-state loop.

Because the inference is based on millivolt-scale telemetry displacement, reconstructed threshold timing and local differential-capacity estimation, the appropriate publication wording is **approximately 60 mA per selected cell**, accompanied by a conservative engineering range of 50–70 mA.

Publication-ready conclusion paragraph

Comparison of two repeated charge cycles at matched cell-7 states indicates an effective passive-balancing current of approximately 60 mA per selected cell. The differential fit corresponds to an effective bleed resistance of approximately 59–61 Ω , or about 58–61 mA over the 3.5–3.6 V upper-cell region, with roughly 0.20–0.22 W dissipation per active channel. The highest cell lost approximately 0.068 Ah during one complete matched-state loop. Because the estimate is indirect and depends on millivolt-scale telemetry and local differential-capacity fitting, a conservative uncertainty range of 50–70 mA should be stated.

A. Per-cell numerical inputs

Table 4 lists the directly calculated voltage displacement and resistor-selection exposure for the high-resolution direct-channel threshold treatment. The exposure column uses the 3.599 V matched loop; the

3.500 V loop differs only in the third decimal place for most cells.

Table 4. Per-cell voltage displacement and balancing exposure. Cell 16 is reconstructed.

Cell	ΔV at 3.500 V (mV)	ΔV at 3.599 V (mV)	T_i (min)	$K_7 - K_i$ (V h)
1	3	5	31.4	2.205
2	3	5	25.8	2.524
3	2	3	57.8	0.688
4	2	3	52.3	1.007
5	2	3	54.3	0.892
6	1	3	0.0	3.989
7	0	0	69.5	0.000
8	1	2	0.0	3.989
9	3	4	33.2	2.102
10	3	4	42.7	1.559
11	0	1	0.0	3.989
12	2	3	0.0	3.989
13	2	4	43.0	1.543
14	3	4	0.0	3.989
15	3	4	18.2	2.956
16*	0	12	18.1	2.962

*Cell 16 is reconstructed as pack voltage minus the sum of cells 1–15 and is not used as a regression observation. Cell 7 is the matched reference cell, so its voltage displacement and differential exposure are zero.

B. Reproducibility procedure and data identifiers

The numerical procedure can be reproduced with the following sequence:

1. Parse the second CSV while ignoring metadata lines beginning with #; construct monotonic sample times from the recorded timestamps.
2. Select the matched sample pairs 965/2329 for cell 7 at 3.500 V and 1103/2467 for cell 7 at 3.599 V.
3. For each sample, calculate $V_{\min}$, evaluate Equation (6), and integrate K_i and T_i with Equations (20) and (21).
4. Estimate each local voltage-time slope by Equation (17) over the eight-minute pre-reference window.
5. Convert each measured voltage displacement to $y_{i,k}$ with Equations (18) and (19).
6. Calculate $x_{i,k} = K_{7,k} - K_{i,k}$ and fit R_{eff} with Equation (25).
7. Repeat with direct-cell and all-cell definitions of $V_{\min}$, and repeat with several local slope-window lengths as a sensitivity check.

Table 5. Source files and SHA-256 identifiers.

File	SHA-256
<code>dyness-balancer-2026-08-09T11-28-32(1).csv</code>	4e732e6e3cf427f77d5684ae96684e9a4858f82f18dc69157a802b0d1180b1c9
<code>dyness-balancer-2026-08-09T11-28-32(2).csv</code>	46edade0770de20567345ceb08af1bead9e6ae5b1fb6cd01aac5ed6ab91f3540
<code>Dyness_BMS_Balancing_Algorithm_Charge_Control_Firmware_2.50-71.10.11-4.pdf</code>	c9a0a458a3211e8d9c320afcacd5189a7630be4becee3843111dca443bb918eb

C. Reference

- [1] Gerdes, H. *Observed Dyness Powerbrick Pro 14.3 kWh BMS and Charge-Control Behaviour: Experimental characterization of passive balancing, full-charge detection, CCL control and charge-MOSFET protection*. Battery software version 2.50-71.10.11, experimental report, 9 August 2026. Reserved DOI: 10.5281/zenodo.21860970.